AIoT Course - Project 2 (Human Gesture Recognition Project)

In this project, you are prompted to create an end-to-end Artificial Intelligence of Things (AIoT) procedure in order to recognize a set of gestures automatically. This problem is identified as Human Gesture Recognition (HGR), which is actually the technology that uses sensors to read and interpret hand gestures as commands.

Key Project Details

- **Announcement date**: 20 April, 2026
- **Delivery Date**: 25 May, 2026, 23:59
- **Team Structure**: A team of a max of 3 students is mandatory for each project.
- **Grade Weight**: This project accounts for 40% of your grade.

The Challenge: Hardware and Data Collection

You will take advantage of the Mbientlab's sensorial device, MetaMotionR research sensor kit, and its wristband. The sensor kit embeds the Bosch BMI160 Inertial Measurement Unit (IMU), which requires highly accurate, real-time sensor kinesiological data.

Using the smart wearable device, you will be recording standard smartphone gestures commonly used while navigating social media. You will collect data for five distinct gestures.

The Gestures:

- **Scroll Up**: The user places their finger on the screen and drags it downward to move the on-screen content up.
- **Scroll Down**: The user places their finger on the screen and drags it upward to move the on-screen content down.
- **Swipe Left**: A horizontal swipe where the finger starts on the right side of the screen and quickly glides to the left.
- **Swipe Right**: A horizontal swipe where the finger starts on the left side of the screen and quickly glides to the right.
- **Texting**: Typing a standard message on the smartphone's virtual keyboard.

Time Requirements:

- Each gesture requires 5 minutes of data collection per subject.
- The total time per subject is 25 minutes.
- The total time per group is 1 hour and 15 minutes, assuming 3 subjects per group.

Machine Learning & Modeling

After collecting the data, you will proceed with data engineering and preparation methodologies in order to transform the data into a suitable format capable of training the AI models. You will select between a set of supervised-learning Machine Learning models for the learning procedure, and finally, you will evaluate the AI models with respect to their performance.

You will train the ML models in two ways:

1. By feeding the ML algorithms with segmented-windowed data.
2. By generating a set of features (feature engineering), and then, feeding the ML models.

Deliverables

You will provide us with a link that leads to a downloadable .zip file (e.g., Google Drive or OneDrive) that will contain:

- The code of the project.
- The collected dataset with its annotated metadata is in the data folder.
- A 2-page report related to the data collection procedure, the classes, how the data was annotated, the controlled environment, how you trained the AI model, and interpreting also the results of the evaluation process.

Further Information

The rest of the information, including setup guidelines, technologies, and reference materials, will be found at the following link:

- https://github.com/AIoT-Group-UoP/IoT-Course-AIoT-project

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